Comparison of Eye-Tracking Representations for Engagement Inference in Office Workers

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Abstract

Eye tracking is an appealing sensor for inferring momentary user states at work. Before training a model, a practitioner must decide how to represent a short gaze window. We compare four representation families on identical three-second windows from an ongoing office-worker study: raw multichannel signals, handcrafted ocular features, frozen GazeMAE embeddings and frozen MOMENT embeddings. The target is a single self-reported item measuring momentary absorption. All four are evaluated under leave-one-subject-out cross-validation with fold-local preprocessing, using a random forest classifier and an MLP against a majority baseline. Raw signals give the highest observed unseen-participant performance (0.538 accuracy, 0.014 above the baseline), handcrafted features follow closely overall, and neither frozen encoder improves on the raw signal. Absolute discrimination is weak throughout (best ROC-AUC 0.551) and between-participant variability dwarfs the differences between representations, so we report the ranking as preliminary, cost-aware guidance.

Keywords

eye tracking, engagement inference, representation learning, GazeMAE, MOMENT, subject generalisation

1 Introduction

Interactive systems that adapt to the person using them need evidence about that person's momentary state. In knowledge work, momentary engagement, particularly absorption in the task at hand, is an attractive target. It fluctuates within a working day and is only weakly visible in application logs. Eye tracking is a plausible sensor for it. Gaze position, fixation and saccade dynamics, blinks and pupil size are available continuously and non-intrusively from a screen-mounted tracker, require no action from the user, and have documented links to attention and processing load.

Turning a stream of gaze samples into model input forces a design decision that is rarely examined on its own, even though benchmark evaluations in time-series classification show that representation families differ substantially in accuracy on the same data [8]. A short window of eye-tracking data can be presented to a classifier in at least four ways. The raw multichannel signal uses selected tracker channels directly, at the price of high dimensionality and little domain structure. Handcrafted ocular features, including fixation, saccade, blink and pupil statistics, encode domain knowledge, stay interpretable, and remain common in applied work [2, 5, 4, 3]. Learned representations promise to remove the manual step. GazeMAE [1] provides gaze-specific embeddings pretrained on eye-movement corpora, while general-purpose time-series foundation models such as MOMENT [6] provide a frozen encoder intended to transfer to unseen domains. The four options differ substantially in preprocessing effort, storage and compute, yet they are seldom compared under the same experimental conditions. Published results differ in windows, labels, classifiers and, most consequentially, in validation protocol, so an apparent advantage of one representation can just as easily come from an optimistic split [9]. We therefore ask which eye-tracking representation supports the most reliable inference of self-reported momentary engagement in unseen office workers. We answer with a controlled comparison of raw signals, handcrafted features, GazeMAE embeddings and MOMENT embeddings on identical three-second windows, an identical binary engagement target and identical leave-one-subject-out folds. Both pretrained encoders are kept frozen and only a downstream classifier is trained, which is the out-of-the-box setting a practitioner would try first. This design provides cost-aware guidance for representation selection under limited labelled data.

2 Related Work

Eye tracking has long been used as behavioural evidence for internal user states. Pupil dilation is a classical correlate of processing load [7], and fixation, saccade, blink and pupil statistics computed over short windows remain common inputs to models of workload, affect and engagement [2, 5]. Our own prior work illustrates the appeal and the limits of this pipeline. Handcrafted ocular features supported coarse affective distinctions [4]. A systematic evaluation across several prediction targets also showed that task type is recognised far more reliably than graded workload, and that reported accuracy depends strongly on how the prediction target and the validation split are defined [3].

Learned representations offer an alternative. GazeMAE pretrains temporal autoencoders on raw position and velocity signals and shows that the resulting gaze-specific embeddings support downstream classification through a simple linear head [1]. MOMENT takes a broader approach by pretraining a multichannel encoder for transfer across time-series domains without task-specific pretraining [6]. The value of both encoders for modest, subject-dependent recordings of spontaneous behaviour remains unclear.

Evaluation design is a further concern. Windows drawn from the same person are highly dependent, so random window splits and preprocessing fitted on the full dataset inflate reported performance, an effect documented in detail for other biosignal domains [9]. Comparisons across representations are interpretable only when every candidate is evaluated on identical windows and labels under a subject-disjoint protocol with fold-local preprocessing.

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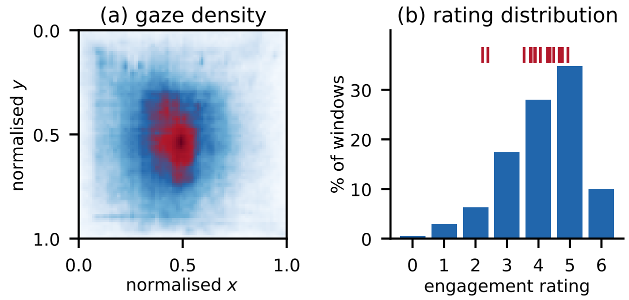

Figure 1: Descriptive context for the modelling cohort. (a) Gaze-coordinate density on the normalised display, pooled over all on-screen samples. (b) Distribution of the engagement rating over the 140,531 modelled windows. The red ticks mark the 17 individual participant means, which span 2.2–4.9 and motivate the participant-aware construction of the target.

3 Methodology

3.1 Data and Eye-Tracking Representations

The data come from a multimodal study in which office workers are recorded during ordinary work at their own workplace. The wider collection records RGB video, eye tracking, microphone audio and screen activity. This paper uses the eye-tracking modality only. Data collection is ongoing and the dataset is not yet public. This preliminary study uses the cohort available at the time of writing, which includes 17 participants and 657 full-workday recordings, with 19 to 79 recordings per participant (median 32). The tracker samples at 60 Hz, and we retain five channels: left and right pupil size, gaze x and y, and the average eye–screen distance.

Labels come from an experience-sampling protocol. Participants are prompted several times per working day with a short questionnaire covering engagement, affect, fatigue, stress, physical comfort and workplace interactions. Our target is the absorption item, "Right now I am immersed in my work", one of three momentary engagement items answered at every prompt and rated on a seven-point scale from 0 to 6. We call it the *engagement rating*. Each prompt labels the 15 minutes of recording that precede it, and only those intervals enter this study.

Recordings are segmented into three-second windows with a 2.25-second step (25 % overlap). Every window carries an identifier built from the participant, the source recording and the position of the window inside that recording. The four representation exports are joined on this identifier, so all models see exactly the same windows. A window is kept only when at least 30 % of its frames are valid and at least 32 valid frames remain, and gaps inside a retained window are linearly interpolated. Of roughly 7.3 million recorded windows, only those inside a labelled interval have an unambiguous target and are usable. Handcrafted features are available for 177,244 of them, 157,685 also pass the encoder quality control, and 140,531 carry a numeric engagement rating. The final modelling cohort accounts for about 2 % of all recorded windows and contains between 2,136 and 25,447 windows per participant.

Display resolutions range from 1920×1080 to 3840×2160 pixels. Resolution differs between participants, and one participant used two different resolutions across recordings. For the raw representation, gaze coordinates are normalised to $[0, 1]$ using each source recording's declared resolution, and out-of-screen coordinates are retained. Overall, 94.7 % of samples fall on the display, and only those enter the descriptive density map in Figure 1(a), which shows the expected central concentration together with broad coverage of the screen.

Four representations are computed from these windows. Coordinate normalisation is applied only to the raw export because the handcrafted features and embeddings were computed earlier and retained unchanged. (1) **Raw signals** use the five channels, truncating longer windows to 120 samples and padding shorter ones with their final value before flattening them into a 600-dimensional vector. (2) **Handcrafted features** describe a window through the ocular events and signals it contains. Blinks, fixations and saccades are taken from the tracker's event classification, and each event type contributes counts, rates and duration statistics, together with the location and dispersion of fixations and the amplitude and velocity of saccades. The remaining groups summarise pupil size and its rate of change, the eye–screen distance and its dynamics, and the fraction of missing samples in each channel. This gives 181 numeric features per window, which are filtered inside each fold as described in Section 3.2. Window identifiers, timestamps and questionnaire columns travel with the export and are not used as model input. (3) **GazeMAE embeddings** are computed with the public micro–macro autoencoder checkpoints applied to pixel-coordinate gaze position and velocity resampled to 500 Hz, with the 128-dimensional position and velocity codes concatenated into 256 dimensions [1]. (4) **MOMENT embeddings** are calculated with the frozen MOMENT-1-LARGE encoder applied to the five-channel windows before display-coordinate normalisation, with an input length of 512 and mean pooling, giving 1,024 dimensions [6]. Both encoders are used out of the box with no weight updates.

3.2 Experimental Design

Engagement ratings are strongly person-specific, with participant means spanning 2.2 to 4.9 on the 0–6 scale (Figure 1(b)). An absolute threshold would therefore mostly encode participant identity. Inside every fold, the ratings of each training participant are mean-centred separately. The held-out participant's ratings are centred with the unweighted mean of the training participants' means, which avoids using held-out labels to set the threshold. A window is positive when its centred rating exceeds zero. Over the cohort this gives 56.3 % positive windows, ranging from 33 % to 72 % per participant.

The primary protocol is leave-one-subject-out cross-validation over the 17 participants. Everything data-dependent is fitted inside the fold. Handcrafted-feature filtering is estimated on the training participants and then applied with unchanged parameters to the held-out participant. For every representation, median imputation and standardisation are fitted on the training participants only. The filtering drops columns with more than 60 % missing values, zero variance, or absolute Pearson correlation above 0.8 with a retained column, leaving 58–60 of the 181 feature candidates.

Three predictors are run on every representation: a majority baseline, a random forest (300 trees, balanced class weights) and an MLP (1028–512–256–128–64 units, GELU activations, layer normalisation, Adam with learning rate 2×10^{-4} and weight decay 10^{-3}, batch size 1024, at most 200 epochs with early stopping after 15 epochs without improvement). For the MLP, a stratified 20 % split of the outer training windows is used for early stopping.

Table 1: Leave-one-subject-out results for binary engagement inference, averaged over the 17 held-out participants. SD is the standard deviation across participants and Δ is the improvement in accuracy over the majority baseline, paired within folds. Balanced accuracy is relative to 0.500 by construction. Accuracy uses all 17 folds. The remaining metrics use the 16 folds whose test participant is not single-class. M-F1 is macro-F1 and AUC is ROC-AUC. Best predictive score and paired improvement in bold.

	Acc.	SD	Δ	Bal. acc.	M-F1	AUC
Majority baseline	.524	.346	—	.500	.326	.500
Raw signals (600-d)						
Random forest	.527	.147	+.003	.529	.460	.546
MLP	**.538**	.113	**+.014**	**.536**	**.468**	**.551**
Handcrafted features (58–60-d)						
Random forest	.528	.230	+.005	.508	.418	.527
MLP	.531	.145	+.008	.521	.451	.535
GazeMAE embeddings (256-d)						
Random forest	.535	.155	+.012	.509	.441	.513
MLP	.520	.107	−.004	.493	.435	.494
MOMENT embeddings (1,024-d)						
Random forest	.521	.286	−.002	.501	.380	.504
MLP	.507	.274	−.017	.497	.362	.504

A single seed is used throughout. The representation × model grid checks whether the representation ranking is robust across models.

We report accuracy as the primary metric, together with its improvement over the baseline paired within folds, and balanced accuracy, macro-F1 and ROC-AUC as secondary discrimination measures. All metrics are aggregated as unweighted means over participants. One participant self-reported every window below the training centre, so that test fold contains a single class. Therefore, accuracy is defined for all 17 folds and the remaining metrics for 16. As a separate within-participant check, we also run a one-step persistence predictor within recordings, carrying the last observed label forward in 90-window test blocks after at least 300 calibration windows.

4 Results

Table 1 reports the main comparison. Every representation–model combination sits close to the majority baseline. The highest observed result is the raw signal with the MLP, at 0.538 accuracy, 0.014 above the paired baseline, and it is also highest on balanced accuracy (0.536), macro-F1 (0.468) and ROC-AUC (0.551). Handcrafted features give the next-highest balanced accuracy, macro-F1 and ROC-AUC, and improve on the baseline accuracy with both classifiers. GazeMAE reaches 0.535 accuracy with the random forest and falls to 0.520, below the baseline, with the MLP. MOMENT falls below the baseline accuracy with both classifiers, while its balanced accuracy and ROC-AUC remain close to chance.

Raw signals give the highest value on every metric. Handcrafted features rank second on the three secondary discrimination metrics, while GazeMAE has the second-highest accuracy. The paired accuracy changes range from −0.017 to +0.014, much less than the spread across held-out participants, whose accuracy standard deviations range from 0.11 to 0.29. The MLP improves raw and handcrafted inputs but lowers both embedding results. A flexible head may help when a representation already carries useful signal. The effect is small, so this interpretation remains tentative.

Figure 2 shows the same difficulty geometrically. In the t-SNE projections, none of the four representations arranges windows by engagement rating, and the colours are as mixed in the learned embeddings as in the raw signal. Colouring the identical projections by participant is more informative. Raw signals form clear participant-dominated patches, handcrafted features and GazeMAE group more weakly, and the MOMENT projection is close to uniform. The clearest visible structure is therefore person-specific, and the weaker grouping in the frozen embeddings may reflect smoothing by the encoders. Because the projections are exploratory, this interpretation and its link to Table 1 remain conjectures.

As a check on the label series, a one-step persistence predictor is evaluated over 1,434 temporal blocks. Participant-averaged accuracy reaches 0.999, while balanced accuracy reaches 0.960 and macro-F1 reaches 0.965 over the 13 participants with two-class test blocks. Each questionnaire answer is broadcast unchanged to every window of the preceding 15-minute interval (Section 3), so most consecutive window pairs share a label by construction. Because the predictor uses observed labels within participants, its result cannot be compared directly with the unseen-participant results above.

5 Discussion

On this dataset, raw signals give the highest observed scores among the four representations for binary engagement inference in unseen office workers, with handcrafted features providing the strongest overall alternative and no improvement from the two frozen embeddings. The small margins over the majority baseline show that three seconds of gaze does not make momentary self-reported engagement well predictable across people, whichever representation is used. The differences between representations are small, and changing the classifier helps some representations while hurting others.

We interpret the lack of improvement from the frozen encoders as a mismatch between their pretraining data and this task. GazeMAE was pretrained on controlled stimulus-viewing corpora and MOMENT on generic time-series benchmarks. Neither was pretrained for day-long, spontaneous office gaze with its dropouts, idle stretches and heterogeneous displays. These results concern frozen transfer only. The next step is to fine-tune the encoders on office gaze, starting with GazeMAE, whose result depended on the classifier (Section 4).

For practice, raw signals scored best here and offer the simplest input pipeline. They require no feature engineering, pretrained checkpoint or encoder inference pass. Handcrafted features should still be considered. They give the strongest overall alternative, they remain interpretable, and our earlier feature-based analyses found them effective on related eye-tracking targets [4, 3], so they are still worth trying on a similar problem. Our results do not support a frozen general-purpose encoder as the default choice.

Several limitations bound these conclusions. The study uses one unpublished and still-growing dataset with 17 participants. The target is one noisy self-report assigned retrospectively to a 15-minute interval and binarised with a fold-specific centring rule. Overlapping windows and repeated labels within each interval reduce the effective sample size. The results depend on one windowing scheme, one seed, one set of quality-control rules

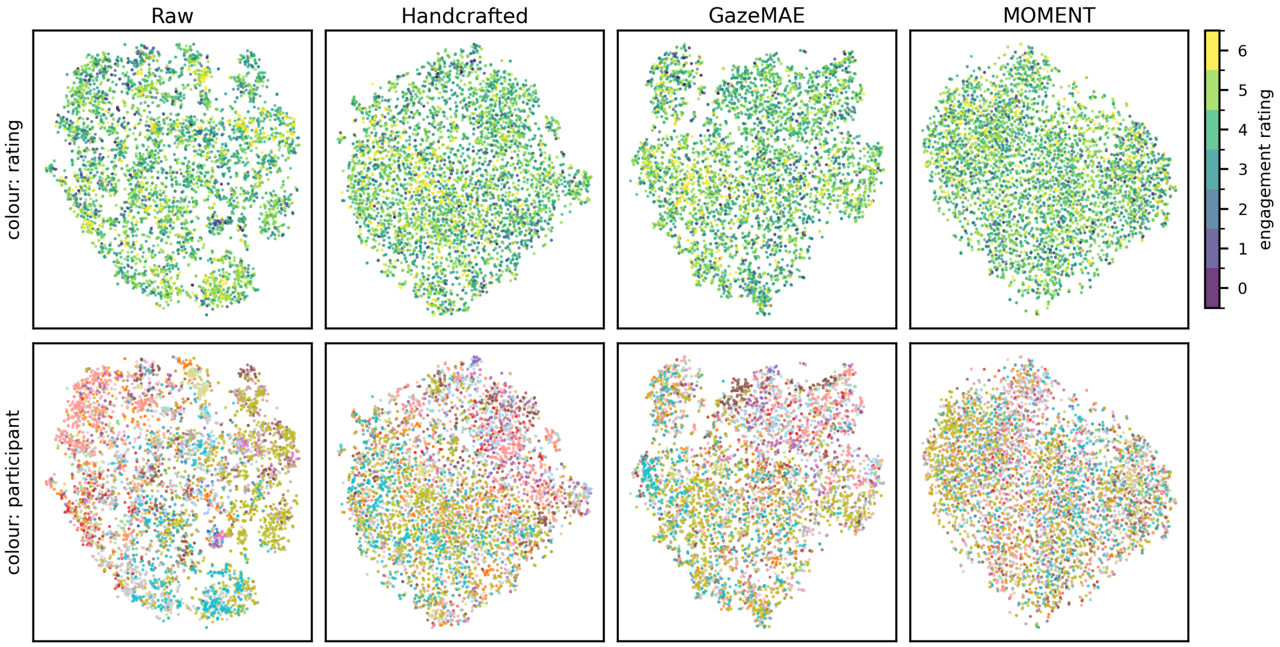

Figure 2: t-SNE projections of the four representations over a common sample of 5,000 windows, coloured by engagement rating (top) and by participant (bottom). Participant colours form visible patches in the raw projection and mix almost uniformly in the MOMENT projection, while no projection shows regions that follow the engagement rating.

and two specific public checkpoints. Coordinate normalisation is applied only to the raw representation, while the earlier feature and embedding exports retain their original preprocessing. The density map and the projections are descriptive only. Future work should validate the comparison on further datasets, explore calibrated personalisation for new users, and test longer windows and aggregated targets that match the granularity at which engagement is actually reported.

6 Conclusion

We compared four eye-tracking representations under one identical, leakage-safe leave-one-subject-out protocol for momentary engagement inference in office workers: raw multichannel windows, handcrafted ocular features, frozen GazeMAE embeddings and frozen MOMENT embeddings. Raw windows produced the highest observed result with 0.538 accuracy and 0.551 ROC-AUC on unseen participants. Out of the box, the frozen encoders did not improve on them, despite their higher preprocessing and compute cost. For practitioners with limited labelled data, raw windows or handcrafted features are the better starting point, while frozen encoders need domain-specific validation or adaptation before use.

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